Numbers

Covering

- Types of number
- Powers and roots
- Fractions, decimals and percentages
- Ordering
- The four operators
- Estimation

Integer
Positive natural numbers and Negative natural numbers with 0 as well, these are Integers.
E.g -92, -11, 0, 5, 13, 412

A prime number is where its only factors are 1, and itself. E.g 2, 3, 19, 101
A composite number is where it has 2 or more factors. E.g 4, 10, 27

Even numbers are numbers which end with 0, 2, 4, 6 or 8
Odd numbers are numbers which end with 1, 3, 5, 7, 9

A square number is the result of a number multiplying by itself. The square root ($\sqrt{\ }$)is the reverse of that. E.g $3^2 = 3 \times 3 = 9$, $\quad \sqrt{9} = 3$, etc
A cube number is the result of a number multiplying by itself three times. The cube root ($\sqrt[3]{\ }$) is the reverse of that. E.g $4^3 = 4 \times 4 \times 4 = 64$, $\quad \sqrt[3]{64} = 4$, etc

A reciprocal of a number is basically 1 over the number, or for a fraction flipping the numerator and denominator.
E.g $3 \rightarrow \frac{1}{3}$, $\frac{3}{4} \rightarrow \frac{1}{\frac{3}{4}} = \frac{4}{3}$

The factors of a number are the numbers which divide evenly.
E.g Factors of 36 are: 1, 2, 3, 4, 6, 9, 12, 18, and 36
Multiples of a number are the numbers in its times table
E.g Multiples of 6 are: 6, 12, 18, 24, 30, ...

The **H**ighest **C**ommon **F**actor between 2 or more numbers is the HCF of them. You select the lowest powers in index form of a number.
The **L**owest **C**ommon **M**ultiple between 2 or more numbers is the LCM of them. You select the highest powers in index form of a number.
Index form is where you write a number as product of primes.

E.g take the HCF and LCM of 840 and 756

$$840 = 2^3 \times 3 \times 5 \times 7$$
$$756 = 2^2 \times 3^3 \times 7$$
$$\text{HCF} = 2^2 \times 3 \times 7 = 84$$
$$\text{LCM} = 2^3 \times 3^3 \times 5 \times 7 = 7560$$

Fractions
Parts of a whole number, they're in the form $\frac{p}{q}$ $(q \neq 0)$. The top is numerator p and bottom is denominator q.
E.g $\frac{1}{2}$, $\frac{3}{4}$, $\frac{-5}{7}$, $\frac{49}{-36}$
These can be written in the form of a decimal number;
Terminating Decimals
These are decimals which stop after a certain number of decimal places.
E.g is $\frac{1}{2} = 0.5$, $\frac{5}{8} = 0.625$, ... because these stop (terminate) after 1 or so decimal places.
Recurring Decimals
These are same decimal numbers but repeat the same or so group of digits.
E.g $\frac{1}{7} = 0.142857142857...$ is recurring because digits 142857 repeat forever. Recurring decimals are written with dots on the first and last digit of the repeating digits. If there is 3 or more digits repeating we use a bar over them.

E.g $0.1\dot{7} = 0.1777...$ $\quad 0.1\dot{2}\dot{3} = 0.1232323...$ $\quad 0.\overline{123} = 0.123123123...$

Recurring Decimals
To be able to convert between a fraction and a recurring decimal, we use multiplying by 10^n, where n is the count of repeating digits.
Let x be the repeating decimal. We subtract $10^n x$ by x and obtain a equation $ax = b$, where we just solve for x, simplify and done.
E.g $0.\dot{3}$, $1.\dot{2}\dot{3}$

$0.\dot{3}$, 1 repeating, hence $\times 10^1 = 10$

$$\begin{aligned} 10x &= 3.333... \\ x &= 0.333... \\ 9x &= 3 \\ x &= \frac{\cancel{3}_1}{\cancel{9}_3} = \frac{1}{3} \end{aligned}$$

$1.\dot{2}\dot{3}$, 2 repeating, hence $\times 10^2 = 100$

$$\begin{aligned} 100x &= 123.232323... \\ x &= \quad 1.232323... \\ 99x &= 122 \\ x &= \frac{122}{99} \end{aligned}$$

Rational Numbers
They are numbers which are of the form $\frac{p}{q}$ $(q \neq 0)$. They are the casual Integers, Terminating Decimals and Recurring Decimals.
Irrational Numbers
They are numbers which can't be written as a fraction. E.g like $\sqrt{2}, \pi$ come under this. They are also under the Non-Terminating Non-Recurring decimals

Percentage
A percent % means $\div 100$ or $\times \frac{1}{100}$. When converting to a percentage, $\times 100$ and append %. When converting from percentage, remove the % and $\div 100$

Estimation
1. Round off every number to the specified degree (usually 1s.f), 2. then compute.
E.g by estimating to 1s.f, compute $\dfrac{3.84 \times \sqrt{0.088}}{2.38}$

$$\begin{aligned} 3.84 &\to 4 \\ 0.088 &\to 0.09 \\ 2.38 &\to 2 \end{aligned} \qquad \frac{4 \times \sqrt{0.09}}{2} = 0.6$$

Ordering

Questions use =, ≠, >, <, ≥, and ≤ to bound or limit answers

The order of operations follow the BODMAS/BIDMAS/PEMDAS rule:

BODMAS	BIDMAS	PEMDAS	
Brackets	Brackets	Parentheses	
Order	Index	Exponent	
Divide	Divide	Multiply	Division and Multiplication are on the same level
Multiply	Multiply	Divide	Hence they can be evaluated from left-to-right
Add	Add	Add	
Subtract	Subtract	Subtract	Same can be said for Addition and Subtraction

Significant Figures

Rules of counting them;

All non-zero numbers are significant: 50702 (3.s.f), 304.005228000 (6s.f), 23.400 (3s.f)

All zeros trapped between non-zeros: $\underline{50702}$ (5.s.f), $\underline{304.005228}000$ (9s.f), $\underline{23.4}00$ (3s.f)

Leading zeros are not significant: ~~000~~$\underline{302}$00 (3.s.f), ~~0.00~~$\underline{4042}$0 (4s.f)

Trailing zeros in a decimal number are significant: $\underline{304.005228000}$ (12.s.f), $\underline{23.400}$ (5s.f), ~~0.00~~$\underline{40240}$ (5s.f)

Finally they are:

50702	304.005228000	23.400	00030200	0.0040240
5s.f	12s.f	5s.f	3s.f	5s.f

Decimal Places

Decimal places mean the count of digits after the decimal point

$$23.\underbrace{40578}_{5d.p} \quad 10.\underbrace{374592}_{6d.p} \quad 0.\underbrace{03}_{2d.p}$$

Rounding

1. When asked upto some significant figures, on a whole number we keep the 0's.
2. When with a decimal, the 0's to the right after the decimal can be ignored.

1. E.g 23542 → 24000 (2s.f) 2. E.g 2.0521 → 2.1[000] (2s.f)

Indices and Standard Form

Covering

- Indices (I)
- Standard Form
- Surds
- Indices (II)

O levels and Extended

Standard Form

A number in the form $a \times 10^n$ where a is a number with decimal after one non-zero digit, from left to right. n is the power of 10. To change a number into standard form, we move the decimal. Moving to the right, the power decreases; Moving to the left, the power increases, by the count you displaced it by.
E.g 587.62, 0.0645, in standard form:

$$5\,\underset{+2}{\underleftarrow{87.}}62 = 5.8762 \times 10^2 \qquad 0.\underset{-2}{\underrightarrow{06}}\,45 = 6.45 \times 10^{-2}$$

Operations with Standard Form

Multiplication and Division is carried out by multiplication and division of the a values, and then using indices rule with powers of 10. Then finally fix the answer once done.

$$3.2 \times 10^5 \times 4.5 \times 10^3 \qquad \frac{6.4 \times 10^7}{8 \times 10^3}$$

$$3.2 \times 4.5 \times 10^{5+3} \qquad \frac{6.4}{8} \times 10^{7-3}$$

$$1\,\underset{+1}{\underleftarrow{5.}}75 \times 10^8 \qquad 0.\underset{-1}{\underrightarrow{8}} \times 10^4$$

$$1.575 \times 10^9 \qquad 8 \times 10^3$$

Addition and Subtraction is carried out by first matching the powers of 10, preferably the smaller power, then adding/subtracting the a values. Then finally fix the answer once done.

$$\left(5.6\times10^4\right)+\left(2.3\times10^5\right) \qquad \left(7.9\times10^6\right)-\left(3.4\times10^5\right)$$

$$\left(5.6\times10^4\right)+\underset{\underset{-1}{\rightarrow}}{2.3}\times10^5 \qquad \underset{\underset{-1}{\rightarrow}}{7.9}\times10^6-\left(3.4\times10^5\right)$$

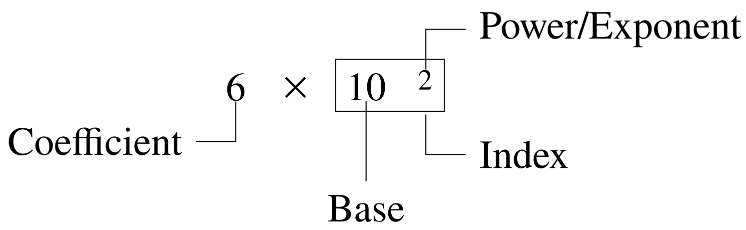
$$(5.6+23)\times10^4 \qquad (79-3.4)\times10^5$$

$$2\,\underset{\underset{+1}{\leftarrow}}{8.6}\times10^4 \qquad 7\,\underset{\underset{+1}{\leftarrow}}{5.6}\times10^5$$

$$2.86\times10^5 \qquad 7.56\times10^6$$

Indices

Indices mean any term raised to some power. The following shows what each word/term means.



Indices Rules

- $a^m\times a^n=a^{m+n}$
- $a^m\div a^n=a^{m-n}=\dfrac{a^m}{a^n}$
- $(a^m)^n=a^{m\times n}$
- $(a\times b)^m=a^m\times b^m$
- $\sqrt[n]{a}=a^{\frac{1}{n}}$
- $a^{-n}=\dfrac{1}{a^n}$
- $a^0=1$
- $\left(\dfrac{a}{b}\right)^n=\dfrac{a^n}{b^n}$
- $\sqrt{a}\times\sqrt{b}=\sqrt{a\times b}$
- $\sqrt{\dfrac{a}{b}}=\dfrac{\sqrt{a}}{\sqrt{b}}$
- $\sqrt[n]{a^m}=\left(\sqrt[n]{a}\right)^m=a^{\frac{m}{n}}$
- $\sqrt[n]{a^n}=\left(\sqrt[n]{a}\right)^n=a$
- $a^n+b^n\neq(a+b)^n$
- $a^n-b^n\neq(a-b)^n$
- $\sqrt{a}=a^{\frac{1}{2}}$
- $\sqrt[3]{a}=a^{\frac{1}{3}}$

Surds **O levels and Extended**

A surd is an irrational number expressed as a square root (or any other). $\sqrt{9}$, $\sqrt{6\frac{1}{4}}$, etc are NOT surds because they can be expressed as rational numbers, i.e 3, $\frac{5}{2}$, etc.
$\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ ARE surds because they aren't rational, cannot be expressed in the form $\dfrac{p}{q}$, $(q \neq 0)$
In the same way, $5+\sqrt{2}$, $5-\sqrt{2}$, $5\sqrt{2}$, $\dfrac{\sqrt{2}}{5}$ are also irrational numbers as.

Simplifying **O levels and Extended**

To simplify a surd, we express it in the form $a\sqrt{b}$ where a is the smallest possible whole number. E.g
Simplify $\sqrt{18}$, $\sqrt{600}$, $\sqrt{448}$, $\frac{\sqrt{8}}{\sqrt{10}}$

$\sqrt{18}$	$\sqrt{600}$	$\sqrt{448}$	$\dfrac{\sqrt{8}}{\sqrt{10}}$
$\sqrt{2\times \underline{3^2}}$	$\sqrt{\underline{2^3}\times 3\times \underline{5^2}}$	$\sqrt{\underline{2^6}\times 7}$	$\dfrac{\sqrt{\underline{2^2}\times 2}}{\sqrt{2\times 5}}$
$\sqrt{2}\times\sqrt{3^2}$	$\sqrt{2^2\times 2}\times\sqrt{3}\times\sqrt{5^2}$	$\sqrt{2^6}\times\sqrt{7}$	$\dfrac{\sqrt{2^2}\times\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}\times\sqrt{5}}$
$3\sqrt{2}$	$10\sqrt{6}$	$8\sqrt{7}$	$\dfrac{2}{\sqrt{5}}$

Rationalizing the denominator **O levels and Extended**

A fraction whose denominator is irrational is inconvenient to work with than it is with a rational denominator. For this reason we often need to rationalize the denominator.

To rationalize the fraction with the denominator of form $b\sqrt{c}$ we just multiply by 1 (so that we don't change the value) using the fact $\frac{\sqrt{c}}{\sqrt{c}} = 1$. E.g

Rationalize the denominators of the fractions: $\frac{2}{\sqrt{5}}, \frac{3}{\sqrt{6}}, \frac{10}{3\sqrt{5}}, \frac{1}{2\sqrt{3}}$

$$\frac{2}{\sqrt{5}} \qquad \frac{2}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \qquad \frac{2\sqrt{5}}{5}$$

$$\frac{3}{\sqrt{6}} \qquad \frac{3}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} \qquad \frac{\sqrt{6}}{2}$$

$$\frac{10}{3\sqrt{5}} \qquad \frac{10}{3\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \qquad \frac{2\sqrt{5}}{3}$$

$$\frac{1}{2\sqrt{3}} \qquad \frac{1}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \qquad \frac{\sqrt{3}}{6}$$

To rationalize the fraction with the denominator of form $a + b$, where a or b or both a and b, are irrational. We use the same technique of multiplying by 1, but a little complex.

The fraction to use is the conjugate of the denominator on numerator and denominator. The conjugate of $a + b$ is $a - b$, and so $a - b$ conjugate is $a + b$.

Recall the difference of two squares, where $(x + y)(x - y) = x^2 \cancel{-xy} \cancel{+xy} - y^2 = x^2 - y^2$.

We use this: $(x + \sqrt{y})(x - \sqrt{y}) = x^2 \cancel{-x\sqrt{y}} \cancel{+x\sqrt{y}} - (\sqrt{y})^2 = x^2 - y$

$(\sqrt{x} + \sqrt{y})(\sqrt{x} - \sqrt{y}) = (\sqrt{x})^2 \cancel{-\sqrt{x}\sqrt{y}} \cancel{+\sqrt{x}\sqrt{y}} - (\sqrt{y})^2 = x - y$

This leaves us with no irrational terms, rationalizing the denominator.

E.g Rationalize the following: $\frac{5}{2 + \sqrt{3}} \quad \frac{\sqrt{7}}{5 - 2\sqrt{3}} \quad \frac{4}{\sqrt{5} + \sqrt{3}}$

$$\frac{5}{2+\sqrt{3}} \qquad \frac{5}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} \qquad \frac{5(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})} \qquad 10 - 5\sqrt{3}$$

$$\frac{\sqrt{7}}{5-2\sqrt{3}} \qquad \frac{\sqrt{7}}{5-2\sqrt{3}} \times \frac{5+2\sqrt{3}}{5+2\sqrt{3}} \qquad \frac{\sqrt{7}(5+2\sqrt{3})}{(5-2\sqrt{3})(5+2\sqrt{3})} \qquad \frac{5\sqrt{7}+2\sqrt{21}}{13}$$

$$\frac{4}{\sqrt{5}+\sqrt{3}} \qquad \frac{4}{\sqrt{5}+\sqrt{3}} \times \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}-\sqrt{3}} \qquad \frac{4(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})} \qquad 2\sqrt{5} - 2\sqrt{3}$$

Everyday Mathematics

Covering

- Ratio and proportion
- Rates
- Percentages
- Money
- Exponential growth and decay

Ratio

Ratio means division, E.g $2 : 3 = \frac{2}{3}$, or parts, E.g 10 in ratio $2 : 3 = 4 : 6$

Direct Ratio

When we increase or decrease one value, the other value also increases or decreases. For this we multiply diagonally and solve for missing. E.g

$$\begin{array}{ccc} 2 & : & 3 \\ 6 & : & x \end{array} \qquad 2x = 18, \qquad x = 9$$

Indirect Ratio

When we increase one quantity, the other quantity decreases, and vice versa. For this we multiply horizontally and solve for missing. E.g

$$\begin{array}{l} 2 \dashrightarrow 3 \\ 6 \dashrightarrow x \end{array} \qquad 6 = 6x, \qquad x = 1$$

Percentage

Percentage means out of 100. To solve questions of the form, **Part** is **Percent** percent of **Whole**, we setup:

$$\text{Part} = \frac{\text{Percent}}{100} \times \text{Whole}$$

Where if anything is ”what”, it means to find that.

Percentage
E.g

$$\text{What is } \textbf{Percent} \text{ percent of } \textbf{Whole}? \qquad \text{What is 12 percent of 25?}$$
$$x = \frac{\text{Percent}}{100} \times \text{Whole} \qquad x = \frac{12}{100} \times 25, \quad x = 3$$

$$\textbf{Part} \text{ is what percent of } \textbf{Whole}? \qquad \text{45 is what percent of 9?}$$
$$\text{Part} = \frac{x}{100} \times \text{Whole} \qquad 45 = \frac{x}{100} \times 9, \quad x = 500$$

$$\textbf{Part} \text{ is } \textbf{Percent} \text{ percent of what?} \qquad 15 \text{ is } \frac{3}{5} \text{ percent of what?}$$
$$\text{Part} = \frac{\text{Percent}}{100} \times x \qquad 15 = \frac{\frac{3}{5}}{100} \times x, \quad x = 2500$$

Percentage Increase
This is used for profit or price increase.

$$\text{Percent Increase} = \frac{\text{Increase Amount}}{\text{Original Whole}} \times 100\%$$
$$\text{Increase Amount} = \text{New Whole} - \text{Original Whole}$$

Remember to use Original Whole in the denominator, it may be the value it used to be/before hand.

Percentage Decrease
This is used for loss/discount or price decrease.

$$\text{Percent Decrease} = \frac{\text{Decrease Amount}}{\text{Original Whole}} \times 100\%$$
$$\text{Decrease Amount} = \text{Original Whole} - \text{New Whole}$$

Remember to use Original Whole in the denominator, it may be the value it is now/after wards.

Ratio Method for Percentage

The Original Whole will always be with 100, the New Whole will either be with $100 + x$ (Increase) or $100 - x$ (Decrease), depending on the question.
The final step is to cross multiply and solve for the unknown, either for Original Whole, New Whole, or $100 \pm x$.

	Amount	**Percent**
Original	Original Whole	100
New	New Whole	$100 \pm x$

$$(\text{Original Whole})(100 \pm x) = 100(\text{New Whole})$$

Interest

Simple Interest is by $I = \dfrac{P \times r \times T}{100}$, where I is Simple Interest, P is Principle Amount or amount borrowed, r is the Interest Rate (in %), T is Time in years.
If R is in months, then T should be in months. Else T should always be in years.
Total Amount = $A = I + P$.

Compound Interest is, Total Amount = $A = P\left(1 + \dfrac{r}{100}\right)^T$.
Compound Interest itself is C or $I = A - P$.

Scale and Maps

In short, the test papers have the smaller length while the actual length is the larger one. If no unit is given, assign both to cm and convert the larger into the unit of actual length.
E.g If we got 1:40000, assign cm 1 cm : 40000 cm, convert 1 cm : 0.4 km

In case of $1 : n$, assign a unit and divide till you reach the same length unit in the actual map, then by comparison find value of n.
E.g Map scale is 1:250000, find distance on the map if the actual distance is 125 km
Assign, 1 cm : 250000 cm, Convert cm 1 cm : 2.5 km, Compare

$$\begin{aligned} 1 \text{ cm} &: 2.5 \text{ km} \\ x \text{ cm} &: 125 \text{ km} \\ x &= 50 \text{ cm} \end{aligned}$$

In case of area, get the square of the lengths by squaring both sides, then compare.
E.g 1 cm : 5 km, find actual area for 1 cm^2 on the map
$(1 \text{ cm} : 5 \text{ km})^2 = 1 \text{ cm}^2 : 25 \text{ km}^2$

Scale and Maps

In case of scale and length, always apply ratio method to find missing.

E.g On a map with scale 1 : 50000, a wall is 6 cm long. How long will it be on a different map with scale 1 : 30000.

Map A; Assign, Convert 1 cm : 0.5 km. Map B; Assign, Convert 1 cm : 0.3 km.

Compare 6 cm : 3 km. 3 km is the actual distance, so we need to find what length is 3 km on map B 1 cm : 0.3 km

x cm : 3 km

x= 10 cm.

Conversion Factors

Length	Volume	Mass
• 1 **k**ilo**m**etre = 1000 **m**etre	• 1 ℓitre = 1000 cm^3	• 1 tone = 1000 **k**ilo**g**ram
• 1 m = 100 **c**enti**m**etre	• 1 m^3 = 1000 ℓ	• 1 kg = 1000 **g**ram
• 1 cm = 10 **m**illi**m**etre	• 1 **k**iloℓitre = 1000 ℓ	• 1 g = 1000 **m**illi**g**ram
	• 1 ℓ = 1000 **m**illiℓitre	

Exchange Rate

Basic money conversion, question will provide the rates and use ratio method to solve.

E.g Person exchanges 500 USD into JPY when the exchange rate is USD = 150 JPY.

$$\begin{array}{ccc} \text{USD} & & \text{JPY} \\ 1 & : & 150 \\ 500 & : & x \end{array}$$

$$x = 75000$$

Exponential growth and decay **O levels and Extended**

The increase in population/quantity over a period of time at some rate will have the equation.

$$u\left(1+\frac{r}{100}\right)^t = v \quad \text{Initial}\left(1+\frac{\text{rate}}{100}\right)^{\text{time}} = \text{Final}$$

The decrease in population/quantity over a period of time at some rate will have the equation.

$$u\left(1-\frac{r}{100}\right)^t = v \quad \text{Initial}\left(1-\frac{\text{rate}}{100}\right)^{\text{time}} = \text{Final}$$

Time

Covering

- Time

Time

24-hour time is written in four digits. The first two digits represents hours, the last two digits represent minutes.
12-hour time is written as a decimal, whole is hour and decimals are minutes, and then a.m. or p.m.
E.g 21 34 = **21:34** or **9:34 P.M.**
7.09 a.m = **7:09** or **7:09 A.M.**

To change 12-hour time into 24-hour time, we:
If the time is in **A.M.**, remove ”A.M.” and write them in four digits
If the time is in **P.M.**, add **12:00** to the given time and remove ”P.M.”
E.g 7:45 A.M. → **07 45**
7:45 P.M. → 7:45 + 12:00 = 19:45 → **19 45**

To change 24-hour time into 12-hour time, we:
If the time is less than 12:00, put **A.M.**
If the time is more than 12:00, subtract **12:00** and put **P.M.**
E.g 07 45 → **7.45 A.M.**
19 45 → 19:45 - 12:00 = 7:45 → **7.45 P.M.**

Noon means **1200** or **12.00 P.M.**, Midnight means **00:00** or **12.00 A.M.**

If the time is more than **24:00**, subtract **24:00** and put ”next day” with the answer.
If the time is next day, add **24:00** to get today time.

$$\text{E.g}\quad \begin{array}{r l} 23{:}30 & \text{Time started} \\ +1{:}20 & \text{Movie length} \\ \hline 24{:}50 & \text{Time ended} \end{array}$$

24:50 > 24:00 → **00 50 next day**

The flight took 2 hours and the plane landed at 01 34 the next day.

0134 next day + 24:00 = **2534** today.

$$\begin{array}{r l} 25{:}34 & \text{Landed} \\ -2{:}00 & \text{Flight length} \\ \hline 23{:}34 & \text{Departed} \end{array}$$

the plane took off at **23 34**

Addition and Subtraction of time

Add or Subtract hours and minutes separately.
If minutes is more or equal to 60, add 1 to hour and remove 60 from minutes. To borrow minutes during subtraction, borrow 1 hour (60 minutes) and add 60 to the minutes to continue subtraction.
E.g

4 hr 50 min + 3 hr 25 min= $\begin{array}{r} 4{:}50 \\ +3{:}20 \\ \hline 7{:}75 \end{array}$ 75 min → 1 hr 15 min. 7h + 1 hr 25 min = **8 hr 15 min**

7 hr 20 min - 4 hr 45 min= $\begin{array}{r r} & 60 \\ \overset{6}{\cancel{7}}{:} & \cancel{2}^{1}0 \\ -4{:} & 45 \\ \hline 2{:} & 35 \end{array}$ = **2 hr 35 min**

To convert hours into minutes, multiply hours by 60. For decimal, multiply by 60 to get it in minutes
E.g 3.75 hr, .75 hr × 60 = 45 min, = 3 hr 45 min

To convert minutes into hours, divide minutes by 60.

Journey

Arrival Time ($A.T$) = Departure Time ($D.T$) + Journey Time ($J.T$)
Departure Time and Journey Time can just be solved for.
Journey Time is always in hour and minutes.

Limits of Accuracy

Covering

- Limits of Accuracy

Upper bound and Lower bound

Questions start with a given value, which is to an estimated unit amount, e.g nearest 10 cm, 1 m, etc.
Upper bound or greatest value is the highest this "estimated" value could be.
It is, G_U = Given value + $\frac{1}{2}$(Estimated Value)
Lower bound or lowest value is the lowest this "estimated" value could be.
It is, G_L = Given value − $\frac{1}{2}$(Estimated Value)
The estimated unit must be same, if not then make them the same unit before computing. Make them the unit of Given value.

Operations on upper and lower bounds **O levels and Extended**

For this the upper and lower bound are applied before using them. Always find them before doing so. Here the notation is X_U is upper bound and X_L is lower bound of value X.

Upper Bound Lower Bound

Addition

$$C_U = A_U + B_U \qquad C_L = A_L + B_L$$
$$C_U = A_U + B_U + D_U + \dots_U \qquad C_L = A_L + B_L + D_L + \dots_L$$

Subtraction

$$C_U = A_U - B_L \qquad C_L = A_L - B_U$$
$$C_U = A_U - B_L - D_L - \dots_L \qquad C_L = A_L - B_U - D_U - \dots_U$$

Multiplying

$$C_U = A_U \times B_U \qquad C_L = A_L \times B_L$$
$$C_U = A_U \times B_U \times D_U \times \dots_U \qquad C_L = A_L \times B_L \times D_L \times \dots_L$$

Division

$$C_U = \frac{A_U}{B_L} \qquad C_L = \frac{A_L}{B_U}$$
$$C_U = \frac{A_U}{B_L \times D_L \times \dots_L} \qquad C_L = \frac{A_L}{B_U \times D_U \times \dots_U}$$

Set Language and Notation

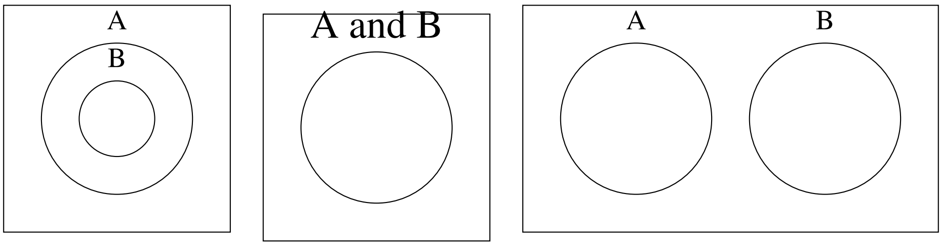
Covering

- Sets

Set Language

Symbol	Definition	Use/Explain
n(A)	Number of elements in set A	A={1,2,3,4} n(A)=4
$\in$	'... is an element of ...'	The left of $\in$ is an element of the set on the right. E.g $4 \in B$ means 4 is an element of B
$\notin$	'... is not an element of ...'	Opposite to $\in$, tells the left side of $\notin$ is not an element of the set on the right
A'	Complement of A	Everything BUT elements of A, $\xi - A$ E.g $A = \{1, 2, 3\}, \xi = \{1, 2, 3, 4, 5\}, A' = \{4, 5\}$
$\emptyset$	The empty set	{}, a set with nothing
ξ	Universal set	Has every element
$A \subseteq B$	A is a sub set of B	Where every element of A is inside B. E.g $A = \{A, E, I\},\ B = \{A, E, I, O, U\},\ A \subseteq B$
$A \subset B$	A is a proper sub set of B	Where every element of A is inside B, but not equal to B
$A \nsubseteq B$	A is not a sub set of B	Implies the opposite of $\subseteq$, where every element of A is not inside B, or A has elements outside of B. E.g $A = \{A, R, I\},\ B = \{A, E, I, O, U\},\ A \nsubseteq B$
$A \cup B$	A union B	All elements of A and B but write common elements only one time. E.g $A = \{1, 2, 3\},\ B = \{2, 3, 5, 7\}$ $A \cup B = \{1, 2, 3, 5, 7\}$. 2,3 common but written once
$A \cap B$	A intersection B	All elements of A and B which are only common. Like how union write them once, we write only them. E.g $A = \{1, 2, 3\}, B = \{2, 3, 5, 7\}, A \cap B = \{2, 3\}$

$\subset$ and $\subseteq$ | $\subseteq$ | Disjoint sets

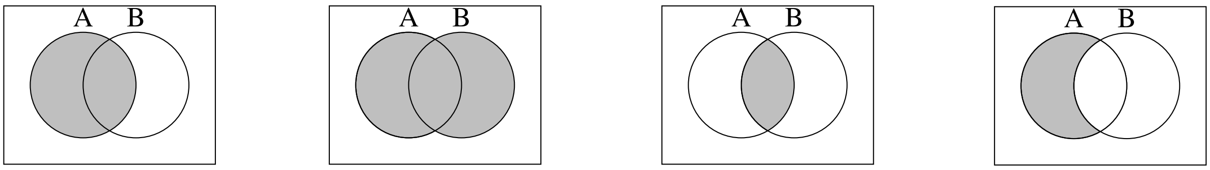


Disjoint Sets don't have anything in common, such $A \cap B = \emptyset$

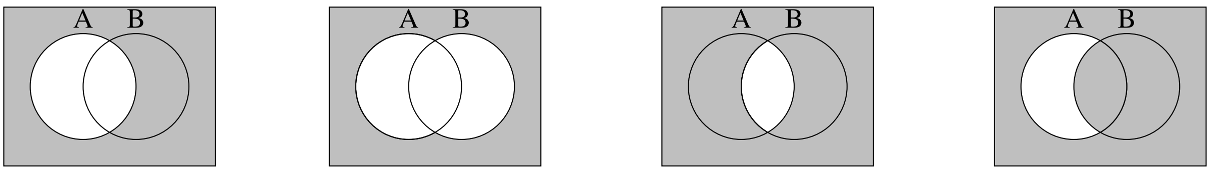
Set Language
There are 2 types/ways questions will provide the contents of a set.
Tabular (or Roster, List, etc) Notation/Form. It is all the elements listed in the set separated by commas. Dots can can be used to indicate repeating sections.
E.g $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6, ..., 20\}$
Set Builder (or Rule, Property, etc) Notation/Form. You describe the contents by a rule, condition, etc. Usually uses a | or : meaning "such that". Generally its $A = \{x :$ condition $\}$
E.g $A = \{x : x \text{ is a prime number}\}$, $B = \{x : x \text{ is even}\}$

Shading
The list represents what part represents what expression. The expression represents to shade that area.
The general method is: $A \cup B$ means to shade expression A AND expression B. $A \cap B$ means to shade the overlap/double shaded regions of what expressions A and B suggest.

A | $A \cup B$ | $A \cap B$ | $A \cap B'$



$(A)' = A'$ | $(A \cup B)' = A' \cap B'$ | $(A \cap B)' = A' \cup B'$ | $(A \cap B')' = A' \cup B$

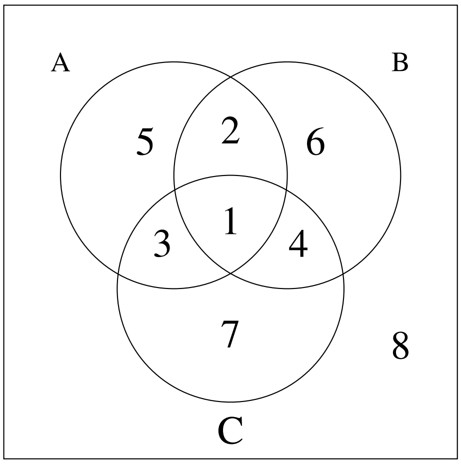


De Morgan Law

The second row uses De Morgan Law, it states the complement of an expression, say E, is basically switching $A \to A'$, $\cup \to \cap$, and vice versa. For shading it means to shade the opposite of what the expression shades

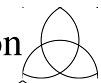
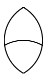
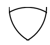
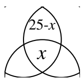
Some expressions representing labelled areas.

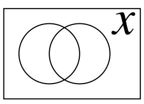
1. $(A \cap B) \cap C$
2. $(A \cap B) \cap C'$
3. $(A \cap C) \cap B'$
4. $(B \cap C) \cap A'$
5. $(B \cup C)' \cap A$
6. $(A \cup C)' \cap B$
7. $(A \cup B)' \cap C$
8. $((A \cup B) \cup C)'$



Completing a Venn Diagram

To complete a Venn Diagram, the process is:

1. Start from the inner most part, i.e $A \cap B$ or $A \cap B \cap C$.
2. To write next value in the Venn Diagram, subtract all previous values from given values. E.g in this section  if this  has the value 25, we minus the value inside , say x and put it in this . So we get 

3. If there's **only** written, then don't subtract any value from it. Write it in its corresponding part of the diagram.
   E.g If we have people who plays **only** football, then that count would go into  area.
4. For **neither** or **nor**, etc, write that value outside any circle but within the universe.
   E.g If we have people who don't read Books nor Magazines, count being x, it will be placed outside the circles 

5. If no value is given or some section, assign it any variable like x, y, z. After assigning all parts of Venn Diagram make equations and solve for them.
   E.g How I did in point 2, assigning  a value as its not given

Using a calculator

Covering

- Using a calculator

This is NOT a thing notes don't contain. This is just guidance I believe everyone should know which comes useful when attempting $\overset{\text{O levels}}{\text{Paper 2}}$ or $\overset{\text{IGCSE}}{\text{Paper 1 or Paper 3.}}$ of the syllabus. I recommend using the $fx{-}991ES\ PLUS$ Casio calculator, it contains everything you need.

Numbers

You need to know how to use the $\frac{\blacksquare}{\square}$ fraction button. You navigate using the big circle D-pad. This way fraction computation is easier as you can see what you're doing.

E.g computing $\frac{2}{3} + \frac{\sqrt{5}}{2}$, $\boxed{\frac{\blacksquare}{\square}, 2, \downarrow, 3, +, \frac{\blacksquare}{\square}, \sqrt{\blacksquare}, 5, \downarrow, 2} = \frac{4+3\sqrt{5}}{6}$ $\boxed{S \Leftrightarrow D} = 1.784...$

$\blacksquare\frac{\square}{\square}$ or mixed fraction button is useful for mixed fractions questions. Technically it means to add the whole number with the fraction, which is what mixed fraction is. It is visually different with the mixed number to the front $_{\text{Being small}}$ like its inside a power, else things like $2\frac{2}{2}$ $\boxed{\frac{\blacksquare}{\square}, 2, \downarrow, 2, \leftarrow, \leftarrow, \leftarrow, \leftarrow, 2}$ is different, meaning $2 \times \frac{2}{2}$.

$x^{\blacksquare}$ and $\sqrt[\blacksquare]{\square}$ are also useful for Indices question if x^2, x^3, $\sqrt{\blacksquare}$, $\sqrt[3]{\blacksquare}$ can't be used, e.g finding $4^{\frac{1}{4}}$

Algebra and graphs

Doing **MODE**, **5: EQN**, **1:** $a_nX + b_nY = c_n$, we can input the values/coefficients of a simultaneous equation and solve it quickly, showing us the correct answer we need to prove by working.

Doing **MODE**, **5: EQN**, **3:** $aX^2 + bX + c = 0$, we can input the values/coefficients of a quadratic equation and get the solutions x_1, x_2 which can help in hinting at how to factorize the question to get the working marks.

You'd want to get it to $(x - x_1)(x - x_2) = 0$. If either $x1$ or x_2 have a denominator, make that the coefficient of x inside the term and keep the numerator.

E.g we got $(x - \frac{3}{4})(x - 1) = 0$, we can shift the denominator to the coefficient of x which will give us the wanted answer, $(4x - 3)(x - 1) = 0$

Mensuration
Here you can use the value of π to compute the actual answer, to 3s.f or 1d.p. To input π its **SHIFT**, $\times 10^x$

Trigonometry
Using the sin, cos, tan, and their inverses, are used to find the values of lengths, angles.

We need to set our calculators in **Degree mode**, the angle type the calculator assumes (without inputting $^\circ$, ^{r}). We set it by doing: **SHIFT**, **MODE**, **3: Deg**

For the trig functions when using equations like, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$, we need to be careful to put) after the angle.
E.g cases like $\dfrac{5 \times \sin(25)}{\sin(30}$ or $\dfrac{4 \times \sin(70}{3}$ will give us Syntax ERROR because it doesn't know what to use as input when its inside a fraction. So putting) at the end tells clearly what the trig function's argument.

For the inverse functions, they're done by: **SHIFT**, **sin** for $\sin^{-1}$, **SHIFT**, **cos** for $\cos^{-1}$, or **SHIFT**, **tan** for $\tan^{-1}$. The input of the inverse functions must be, for $\sin^{-1}$ and $\cos^{-1}$ between -1 and 1, else you'd result in Math ERROR because sin and cos couldn't output values outside $-1 \leq x \leq 1$ either way.